

What Does Your Short-Answer VQA Score Actually Measure? Evaluator-Dependent Instability in Multimodal Short-Answer Benchmarks

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Abstract

Short-answer VQA benchmarks often mix two distinct aspects of model performance: whether an answer is semantically correct and whether its surface form matches the expectation of the automatic evaluator. We study this mismatch across six vision–language models and six benchmarks, using a semantic judge validated by human annotations (97.6% precision) to audit over 37k rejected answers. A second text-only judge yields consistent benchmark-level mismatch patterns, showing that the observed effect is not an artifact of a single audit model. On text-rich benchmarks, up to 47.7% of rejected answers are judged semantically acceptable despite being marked incorrect by the benchmark evaluator. Evaluator sensitivity varies systematically across answer types, with read-out and multi-span answers being particularly evaluator-sensitive compared with scalar answers. Alternative short-answer prompt formulations further destabilize benchmark outcomes, producing substantial item-level flips for the same image–question pairs. A deterministic CPU-only contract repair further shows that part of this undercount is recoverable.

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
Computer vision problems.

Keywords

visual question answering, benchmark evaluation, vision–language models, answer contracts

ACM Reference Format:

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1 Introduction

A model looks at a photograph of a jet on a runway and is asked “What airline is this?” It outputs “Air France”—two words, correctly capitalized, unambiguously identifying the carrier. The gold answer is “Airfrance”—one word, no space. The benchmark marks it wrong. The error does not reflect a failure of visual understanding; the model read the livery correctly. It is a failure of the evaluation contract: the answer is semantically acceptable, but its surface form does not survive the string-matching filter. Figure 1 shows this case alongside a second example where a highway-sign question is answered correctly in both form and content.

Short-answer VQA benchmarks have driven much of the progress in text-rich multimodal understanding over the past decade [44]. ST-VQA [5] and TextVQA [40] stress scene text, DocVQA [33] and InfographicVQA [32] extend the challenge to documents and infographics, and ChartQA [31] adds visual-logical reasoning over charts. Despite these differences, they share one evaluation habit inherited from the original VQA setup [3]: compare the emitted string against one or more gold references under an automatic metric. Exact match, ANLS [36], and relaxed numeric accuracy differ in detail, but all make a string-level comparison decisive.



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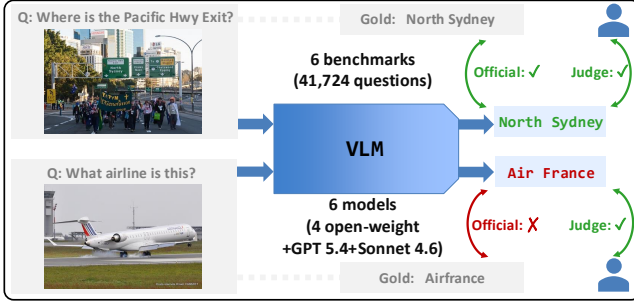


Figure 1: 21–48% of official errors on ST-VQA and TextVQA are semantically acceptable.

String-level evaluation suited an era of short, label-like outputs but fits modern free-form generators less well. Once models answer in natural language rather than from a fixed vocabulary, answer wrapping (“The answer is 42”), morphology, and stylistic variation become part of the measured outcome. Ji *et al.* [18] argue that rigid exact-match patterns penalize correct but differently worded answers, and Ging *et al.* [14] likewise find substantial disagreement among exact-match, relaxed, and judge-based metrics on open-ended VQA.

Recent VQA research has increasingly treated evaluation design itself as part of the measurement problem. Gao *et al.* [12] show that structural biases in the answer space can inflate continual-VQA performance and obscure robustness under distribution shifts. Mai *et al.* [30] further show that broad VQA evaluation can conflate multiple underlying visual abilities and training–evaluation mismatch, making aggregate errors difficult to attribute. In visually rich document VQA, Napolitano *et al.* [35] evaluate whether models can distinguish answerable from plausible but unanswerable questions, while Shaar *et al.* [39] show that evaluation protocol design materially affects model discrimination for open-ended VideoQA. Related work outside VQA has likewise emphasized evaluation and decision criteria that remain aligned with the intended interpretation or downstream use of model outputs [24, 53]. These studies show that VQA scores depend on evaluation design, but they do not isolate failures introduced at the final short-answer matching step.

An official score—say, 84.3% on ST-VQA—does not represent semantic question-answering success alone. It is the share of outputs that survive a benchmark-specific comparison rule applied to a benchmark-specific reference set. Some rejected outputs are genuine failures, but others are semantically acceptable answers that miss the benchmark’s preferred surface form. Our study focuses on this gap between semantic acceptability and benchmark acceptance.

The extent of this gap depends on the form of the expected answer. A numeric question like “How many people?” has a small set of acceptable answer strings: “4”, perhaps “four”, and not much else. A text-reading question like “What does the sign say?” has a much larger set: “STOP”, “Stop”, “The sign says STOP”, “It reads STOP”—all semantically correct, but only some will survive exact match. We call this structural variation the *answer contract*: the implicit agreement between the benchmark and the model about what form an acceptable answer should take. Different answer

contracts—scalar, OCR readout, identifier codes, multi-span lists—create different evaluator sensitivities.

Because official scores can undercount semantic success by different amounts across benchmarks and answer types, adjacent score gaps become harder to interpret, official error pools become contaminated by false failures, and cross-benchmark comparisons stop measuring a common quantity. A model that appears equally strong on two datasets may therefore be passing through two different evaluator filters. Prompt sensitivity compounds the problem: recent work has shown that language models are sensitive to prompt phrasing [9, 28, 38, 49, 55], and we hypothesize that similar instability may emerge in multimodal short-answer evaluation. Even among short-answer prompt pairs with closely matched output lengths, 6–13% of item-level official outcomes can flip while the image and question remain fixed. Evaluator-induced mismatch therefore affects not only absolute scores but also error analysis and cross-benchmark comparison.

- We establish evaluation mismatch as a systematic measurement issue in short-answer VQA through a large-scale study spanning six VLMs, six benchmarks, and over 41k questions. A human-validated semantic audit reveals that benchmark evaluators can reject a substantial fraction of semantically acceptable responses, exposing a non-negligible gap between benchmark acceptance and semantic success.
- We introduce an answer-contract decomposition that characterizes short-answer evaluation by the expected form of the answer, including scalar, readout, identifier-like, and multi-span contracts. This decomposition reveals systematic differences in evaluator sensitivity across answer forms and helps explain why benchmark undercount varies substantially across datasets.
- We reveal substantial evaluation instability under alternative short-answer prompt formulations. Even when the image and question remain unchanged, minor instruction changes can produce pronounced item-level shifts in benchmark-facing outcomes, showing that short-answer VQA scores are sensitive not only to model capability but also to how answers are elicited and evaluated.

2 Related Work

VQA evaluation. String-based evaluation remains common in short-answer VQA, from the original protocol [3] to ST-VQA, TextVQA, DocVQA, InfographicVQA, and ChartQA [5, 31–33, 40]. Although their scoring rules differ, all compare generated answers against pre-defined references. More flexible evaluation has therefore emerged [18, 19, 51], with recent benchmarks extending toward long-form and open-ended responses [41]. We focus on evaluator mismatch at the final answer-comparison interface.

Generative multimodal answering. Large multimodal models have shifted visual-language interaction from fixed-label prediction toward free-form generation [1, 22, 27]. Recent systems further extend this paradigm to structured reasoning, multimodal knowledge-graph completion, and dynamic modality fusion [43, 47, 52]. VQA interfaces are also diversifying: VoQA studies visual-only and OCR-assisted settings [2], while M³-VQA considers different evidence-access conditions [29]. Increasingly flexible outputs and interfaces

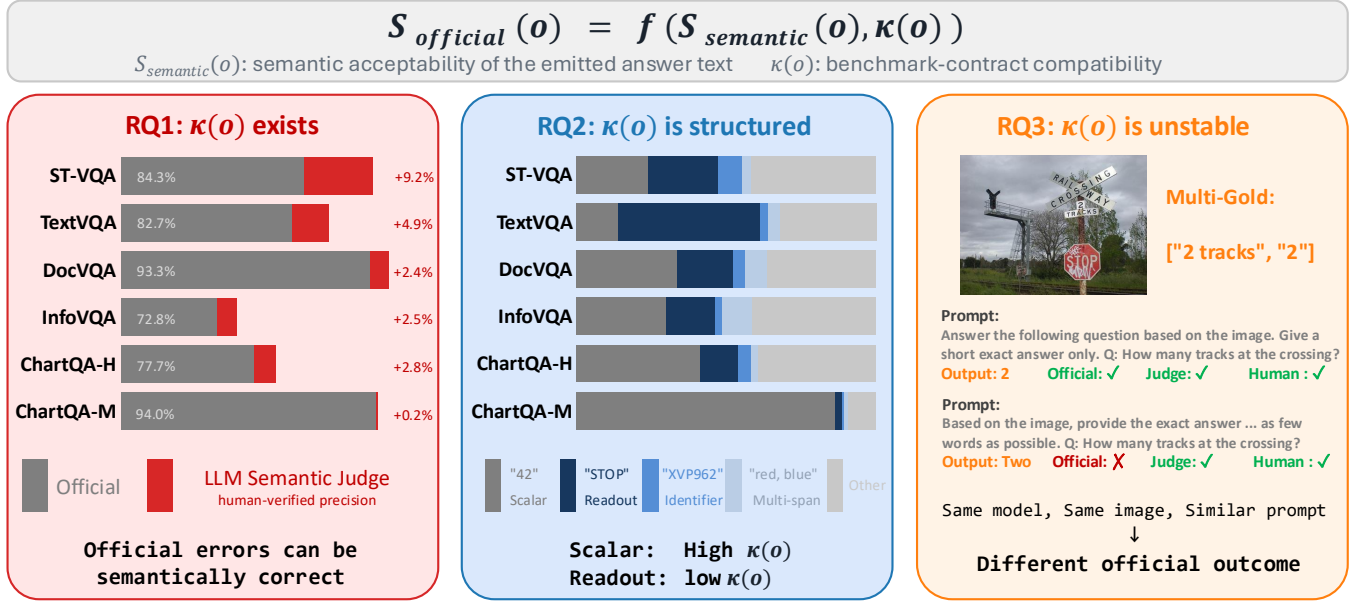


Figure 2: Overview of benchmark undercount, answer-contract structure, and perturbation-induced instability.

make compatibility with a fixed evaluation rule less equivalent to semantic task success.

Evaluation sensitivity to evidence conditions. Measured performance can depend on how evidence is represented, perturbed, or exposed to the model. Related work across multimodal, federated, and robust learning has examined variation under heterogeneous representations, noisy observations, and changing information conditions [10, 11, 16, 25, 48]. Similar effects arise in multimodal QA: misleading chart designs and alternative inference representations can substantially change question-answering performance [42]. Such findings highlight that measured performance reflects not only task competence but also the conditions under which evidence is presented and assessed.

Semantic variation and prompt sensitivity. Free-form generation weakens the correspondence between semantic correctness and a single surface realization. Prior work has studied ambiguous questions, multiple valid answers, one-to-many mappings, and semantic correspondence beyond lexical overlap [8, 17, 34, 37, 46]. Prompt phrasing introduces further variation, with semantically similar instructions producing different outputs [7, 28, 45, 50, 56]. We instead examine how such variation propagates through the evaluator and alters item-level acceptance.

Evaluation mismatch versus hallucination. Evaluator mismatch is distinct from hallucination, which concerns generated content unsupported by the visual input [19, 23]. A benchmark error pool may instead mix genuine failures with semantically acceptable but non-canonical responses. Semantic auditing introduces another uncertainty because model-based judges can exhibit positional, stylistic, and other biases [6, 20, 26, 54]. We therefore retain benchmark-native metrics, validate the semantic judge against human annotations, and use judge decisions only as diagnostic evidence.

3 Framework

3.1 Score Decomposition

For an image I , question q , and prompt condition p , a model produces an output string

$$o = G_\theta(q, I, p). \quad (1)$$

We distinguish two text-level quantities:

- $S_{\text{sem}}(o)$: the semantic acceptability of the emitted answer text;
- $\kappa(o)$: *benchmark-contract compatibility*, i.e. how well the emitted text matches the answer form expected by the official evaluator.

Conceptually, their relationship can be summarized as

$$S_{\text{off}}(o) = f(S_{\text{sem}}(o), \kappa(o)), \quad (2)$$

where S_{off} denotes the benchmark-facing official score and f is a scoring function. The decomposition separates two quantities that benchmark evaluations often conflate: whether the emitted answer is semantically acceptable and whether its surface form satisfies the benchmark’s answer contract. Figure 2 intuitively illustrates several manifestations of $\kappa(o)$, including semantically correct but officially rejected answers, systematic variation across answer contracts, and instability of benchmark-contract compatibility under alternative short-answer prompts.

3.2 Answer Contracts

To quantify how evaluator sensitivity varies across answer forms, we define an operational, rule-based partition by *answer contract*. Items are assigned in priority order using question form, normalized reference shape, and benchmark-native metadata where available, including OCR spans in TextVQA and answer-type annotations in InfographicVQA. The resulting partition contains four named buckets and one heterogeneous residual category, as summarized in Table 1. The categories are explanatory devices rather than a

Table 1: Answer-contract partition.

Bucket	Typical answers and compatibility failures
scalar	Numbers, dates, and yes/no answers
readout	OCR spans, names, and written text
identifier	Codes, room IDs, and alphanumeric strings
multi-span	Lists, conjunctive answers, and multi-span lookups
other	Heterogeneous residual open-form answers

Table 2: Human validation of the text-only judge.

Metric	Value
Human-reviewed items	570
Precision	97.6%
Precision 95% CI	95.3–98.8
Recall	95.5%
Cohen’s κ	0.917

universal ontology of VQA answers. The four named buckets capture recurrent answer forms associated with evaluator sensitivity, whereas *other* contains less internally coherent open-form answers. The partition is used only for post-hoc diagnosis and does not affect model generation or official scoring.

3.3 Undercount Measurement

We estimate benchmark undercount through an LLM-as-Judge semantic audit. In total, we audit 37,033 official-error outputs, including 36,357 from open-weight models and 676 from the evaluated closed-weight model subsets. DeepSeek-Chat is used with a binary *text-only* semantic-equivalence prompt. We intentionally exclude image information because the audit targets evaluator mismatch conditioned on available reference answers, rather than visual reasoning ability.

We validate the judge through a stratified, image-aware human review of 570 official-error items. Human annotators see the image, question, reference answer or answers, and model output. The reviewed items span all six benchmarks, all six models, and all five answer-contract buckets: scalar, readout, identifier, multi-span, and other. As summarized in Table 2, the judge achieves 97.6% precision, 95.5% recall, and Cohen’s $\kappa = 0.917$ against the human-majority labels. Precision remains high across all audit buckets, ranging from 95.1% for other answers to 100.0% for readout and identifier answers.

To test whether the benchmark-level pattern depends on a single judge model, we replicate the full 37,033-item audit using a second text-only judge, moonshot-v1-8k. The two judges show substantial item-level agreement, with Cohen’s $\kappa = 0.760$. Across the six dataset-level observations, their official-error acceptance rates are closely aligned (Pearson $r = 0.999$, $n = 6$), despite different overall acceptance rates of 31.5% and 33.7%. We use these judges as semantic-acceptability diagnostics rather than as replacements for benchmark-native metrics.

4 Experimental Setup

Models. We evaluate four open-weight VLMs: **InternVL3-8B** [57], **Qwen2.5-VL-7B** [4], **Gemma3-12B** [13], and **Llama-3.2-11B-Vision** [15]. We additionally evaluate two closed-weight VLMs, **GPT-5.4** (OpenAI) and **Claude Sonnet 4.6** (Anthropic), on stratified samples: 500 items from each of ST-VQA and TextVQA, and 200 items from each of DocVQA, InfographicVQA, ChartQA-Human, and ChartQA-Machine. The four open-weight models are served with batched vLLM inference [21]. We use deterministic greedy decoding with sampling disabled. Base, A1–A5, and B1–B2 allow up to 24 generated tokens; B3 allows up to 256 tokens because it is used as a no-contract output-length control. Our inference code supplies one user prompt and no system prompt. Each model retains its packaged processor and chat template.

Benchmarks. We evaluate six VQA benchmark settings: **ST-VQA** [5] with 26,074 items, **TextVQA** [40] with 5,000 items, **DocVQA** [33] with 5,349 items, **InfographicVQA** [32] with 2,801 items, and the **ChartQA-Human** and **ChartQA-Machine** splits [31], each with 1,250 items. DocVQA and InfographicVQA use their benchmark-native ANLS evaluation, while both ChartQA splits use 5% relaxed numeric accuracy. We treat ChartQA-Human and ChartQA-Machine as separate benchmarks throughout our analysis because their gold answers are constructed differently—human-authored versus machine-generated—and exhibit distinct answer-contract mixtures. For diagnostic analyses requiring binary outcomes, we use the same frozen benchmark-facing acceptance rule throughout; for DocVQA and InfographicVQA, ANLS scores of at least 0.5 are treated as accepted. Benchmark-native scores are retained where applicable, while these binary labels are used when an item-level accepted/rejected outcome is required.

Perturbation protocol. The RQ3 candidate set contains seven alternative short-answer formulations to Base, together with B3 as a no-contract control. A1–A5 vary the wording of an instruction placed before the question, whereas B1–B2 place alternative short-answer instructions after the question. B3 removes the explicit short-answer contract entirely. The exact user-message templates are:

Base: “Answer directly in one word or phrase, as short as possible. Question: {question} Answer:”

A1: “Look at the image and answer the question below. Provide a brief, exact answer. Question: {question} Answer:”

A2: “Using the provided image, answer the following question with a short exact response. Question: {question} Answer:”

A3: “Based on the image, provide the exact answer below in as few words as possible. Question: {question} Answer:”

A4: “Examine the image and answer the question. Reply with only the short exact answer. Question: {question} Answer:”

A5: “Refer to the image to answer the following. Give only the precise short answer. Question: {question} Answer:”

B1: “{question} Answer in one word or number.”

B2: “{question} Give only the final short answer.”

B3: “{question}”

Table 3: Per-model official and judge benchmark accuracy.

Dataset	Metric	InternVL3	Qwen2.5	Gemma3	Llama-3.2	Sonnet [†]	GPT-5.4 [†]	Open $\bar{\Delta}$	Closed $\bar{\Delta}$
ST-VQA	Official	84.3	75.8	71.6	70.2	61.4	76.4		
	Judge	91.8	85.6	82.4	78.7	69.4	85.8		
	Δ	+7.5	+9.9	+10.8	+8.5	+8.0	+9.4	+9.2	+8.7
	Acc. Err.	47.7	40.8	38.1	28.5	20.7	39.8		
TextVQA	Official	82.7	84.3	76.6	84.0	82.0	82.6		
	Judge	87.5	89.4	82.8	87.4	86.6	88.0		
	Δ	+4.9	+5.1	+6.1	+3.4	+4.6	+5.4	+4.9	+5.0
	Acc. Err.	28.1	32.2	26.2	21.2	25.6	31.0		
DocVQA	Official	93.3	93.8	87.4	91.9	88.5	95.5		
	Judge	94.6	96.7	89.8	94.8	92.0	99.0		
	Δ	+1.4	+2.9	+2.4	+2.9	+3.5	+3.5	+2.4	+3.5
	Acc. Err.	20.2	46.1	18.9	36.0	30.4	77.8		
InfoVQA	Official	72.8	80.1	57.2	68.4	58.0	95.0		
	Judge	75.2	82.0	60.9	70.7	62.0	97.5		
	Δ	+2.3	+1.8	+3.7	+2.3	+4.0	+2.5	+2.5	+3.2
	Acc. Err.	8.5	9.2	8.6	7.2	9.5	50.0		
ChartQA-H	Official	77.7	78.4	63.4	55.3	90.5	93.5		
	Judge	79.9	81.4	66.2	58.2	92.0	95.5		
	Δ	+2.2	+3.0	+2.8	+3.0	+1.5	+2.0	+2.8	+1.8
	Acc. Err.	10.0	14.1	7.6	6.6	15.8	30.8		
ChartQA-M	Official	94.0	94.5	88.2	91.4	88.5	96.5		
	Judge	94.2	94.5	88.7	91.5	92.0	97.5		
	Δ	+0.2	+0.0	+0.5	+0.1	+3.5	+1.0	+0.2	+2.2
	Acc. Err.	4.0	0.0	4.1	0.9	30.4	28.6		

[†] Closed-weight entries come from stratified benchmark samples rather than the full open-weight evaluation pools.

5 Results

We organize our results around four research questions:

RQ1. Undercount: How prevalent and how large is benchmark undercount after semantic audit?

RQ2. Structure: Does evaluator sensitivity vary systematically across answer contracts?

RQ3. Perturbation sensitivity: How often do alternative short-answer instructions change benchmark-facing item-level acceptance outcomes?

RQ4. Contract repair: How much benchmark undercount can deterministic CPU-only contract repair recover?

Unless otherwise stated, *Official* refers to the frozen benchmark-facing acceptance score used in our diagnostic analyses, *Native* refers to the dataset-specific automatic score where reported, and *Judge* refers to the text-only semantic audit applied only to the Official error pool.

5.1 Benchmark Undercount (RQ1)

We estimate $\kappa(o)$ by auditing outputs rejected under the frozen benchmark-facing acceptance rule and identifying those judged semantically acceptable. Table 3 reports, for each model and benchmark, the official score (Official), the corresponding LLM judge score (Judge), their difference (Δ), and the fraction of official errors accepted by the LLM judge (Acc. Err.). Throughout the following analyses, an *official error* denotes an output rejected by this frozen benchmark-facing acceptance rule.

Among the four open-weight models evaluated on the full benchmark sets, the magnitude of undercount varies substantially across benchmarks. The effect is largest on ST-VQA, where 28.5–47.7% of

official errors are judge-accepted and the resulting score correction ranges from +7.5 to +10.8 pp. TextVQA shows a smaller but still substantial effect, with 21.2–32.2% judge-accepted errors and corrections of +3.4 to +6.1 pp. On DocVQA, 18.9–46.1% of official errors are judge-accepted, while the corresponding score corrections are smaller (+1.4 to +2.9 pp), consistent with the tolerance introduced by ANLS-based matching. The corrections are more moderate on InfographicVQA (+1.8 to +3.7 pp) and ChartQA-Human (+2.2 to +3.0 pp). By contrast, ChartQA-Machine exhibits little undercount: only 0–4.1% of official errors are judge-accepted, yielding corrections of at most +0.5 pp.

The stratified closed-weight evaluations show the same overall direction. Both Claude Sonnet 4.6 and GPT-5.4 receive positive judge corrections on all six benchmarks, indicating that evaluator-induced undercount is not specific to the open-weight models.

5.2 Contract Structure (RQ2)

We next examine whether $\kappa(o)$ exhibits systematic structure across answer contracts. Let $\mathcal{I}_b$ denote the set of model–question evaluation rows assigned to contract bucket b under the Base condition, and let $N_b = |\mathcal{I}_b|$. For each row $i \in \mathcal{I}_b$, let $y_i^{\text{off}} \in \{0, 1\}$ denote the frozen Official acceptance, $s_i^{\text{nat}} \in [0, 1]$ the score assigned by the dataset-native automatic evaluator, and $y_i^{\text{judge}} \in \{0, 1\}$ the semantic-judge decision. We focus on the four named contract buckets; the heterogeneous *other* category is excluded from this contract-level comparison. We quantify each contract using three complementary statistics.

First, we quantify the difference between dataset-native automatic scoring and binary Official acceptance. For contract bucket b ,

the Native–Official gap is defined as

$$\Delta_b^{N-O} = \frac{1}{N_b} \sum_{i \in I_b} s_i^{\text{nat}} - \frac{1}{N_b} \sum_{i \in I_b} y_i^{\text{off}}. \quad (3)$$

A positive Δ_b^{N-O} indicates that the dataset-native evaluator assigns more credit on average than the binary Official acceptance, whereas a negative value indicates the reverse. We report Δ_b^{N-O} in percentage points (pp).

Second, we measure the fraction of Official errors that are nevertheless judged semantically acceptable. The judge-acceptance rate among Official errors is defined as

$$\rho_b^{\text{JO}^-} = \frac{\sum_{i \in I_b} \mathbf{1}[y_i^{\text{off}} = 0 \wedge y_i^{\text{judge}} = 1]}{\sum_{i \in I_b} \mathbf{1}[y_i^{\text{off}} = 0]}. \quad (4)$$

Thus, $\rho_b^{\text{JO}^-}$ is conditioned on rejection by the Official acceptance rule rather than on all evaluation rows in bucket b . We report this quantity as a percentage.

Finally, we summarize prompt-induced changes in dataset-native scores across the tested prompt conditions. Let p_0 denote the Base prompt, let $\mathcal{P}$ denote the tested conditions excluding Base, and let $s_{i,p}^{\text{nat}}$ denote the dataset-native score for row i under prompt condition p . The mean signed native-score shift for contract bucket b is defined as

$$\Delta_b^{\text{pert}} = \frac{1}{|\mathcal{P}|N_b} \sum_{p \in \mathcal{P}} \sum_{i \in I_b} (s_{i,p}^{\text{nat}} - s_{i,p_0}^{\text{nat}}). \quad (5)$$

A negative Δ_b^{pert} indicates that the tested prompt conditions reduce the dataset-native score on average relative to Base.

Table 4 reveals systematic differences across answer contracts. Readout answers exhibit a positive Native–Official gap of +3.3 pp and a high judge-acceptance rate among Official errors of 36.4%. Multi-span answers show the highest judge-acceptance rate among Official errors (39.2%), while identifier-like answers exhibit the largest Native–Official gap (+4.0 pp). Scalar answers show a different pattern, with a negative Native–Official gap (−1.8 pp) and a lower judge-acceptance rate among Official errors (21.1%). The native-score shifts provide a complementary view: average scores decrease across the tested prompt conditions for all four named contracts, with Δ_b^{pert} ranging from −9.5 to −11.9 pp. Readout and multi-span answers exhibit the largest mean reductions (−11.9 and −11.8 pp, respectively), followed by scalar answers (−10.7 pp) and identifier-like answers (−9.5 pp). The signed shifts summarize changes in average benchmark score rather than the item-level instability examined in RQ3.

As shown in Table 5, contract composition varies markedly across benchmarks. ChartQA-M is dominated by scalar questions (86.4%), while readout questions account for 47.4% of TextVQA; ST-VQA also contains sizeable readout and identifier-like subsets. Differences in contract composition are consistent with the benchmark-level mismatch patterns observed in RQ1. The partition is additionally supported by benchmark-native annotations: TextVQA provides OCR token structure, and InfographicVQA distinguishes single-span from multi-span answers. The benchmark-native annotations align with our contract categories, suggesting that benchmark-level variation partly arises from differences in answer form.

Table 4: Contract-level evaluation sensitivity.

Bucket	Native–Official	Judge-Accepted Err.	Δ_b^{pert}
Scalar	−1.8 (pp)	21.1%	−10.7 (pp)
Readout	+3.3 (pp)	36.4%	−11.9 (pp)
Identifier	+4.0 (pp)	21.4%	−9.5 (pp)
Multi-span	+2.0 (pp)	39.2%	−11.8 (pp)

Table 5: Dataset-by-contract composition.

Dataset	Total	Scalar	Readout	Ident.	Multi	Other
ST-VQA	26,074	4,707	6,110	2,022	516	12,719
TextVQA	5,000	1,256	2,372	49	119	1,204
DocVQA	5,349	1,396	796	154	770	2,233
InfoVQA	2,801	1,337	113	20	331	1,000
ChartQA-H	1,250	945	24	4	38	239
ChartQA-M	1,250	1,080	21	6	1	142

5.3 Short-Answer Prompt Sensitivity (RQ3)

We next examine whether $\kappa(o)$ is stable across alternative short-answer instructions for the same image–question pairs. All results use the complete benchmark pools. Let $y_i^{(p)} \in \{0, 1\}$ denote the frozen official-acceptance label for question i under prompt condition p , and let N be the number of evaluated questions. For two prompt conditions p_a and p_b , we define the item-level Flip Rate as

$$\text{FlipRate}(p_a, p_b) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i^{(p_a)} \neq y_i^{(p_b)}]. \quad (6)$$

We summarize the fixed short-answer candidate set in two ways. First, A^* is the member of A1–A5 with the largest six-dataset macro-average Flip Rate relative to Base. Second, MaxFlipPair is the pair with the largest macro-average Flip Rate among the 28 unordered pairs formed by {Base, A1, A2, A3, A4, A5, B1, B2}. Table 6 shows that official correctness is not invariant to short-answer prompt wording. The largest Base-to-A-family macro flip is 9.65% for InternVL3, 39.61% for Qwen2.5, 10.44% for Gemma3, and 60.74% for Llama-3.2. Allowing all 28 short-answer prompt pairs raises the corresponding maxima to 10.53%, 41.56%, 21.48%, and 60.74%.

On the other hand, the prompt rewrite may alter the model’s answer realization—for example, switching from an answer fragment to a sentence—and thereby change both semantic behavior and benchmark compatibility. We therefore separate large output-length shifts from comparisons in which average answer length remains similar. For this analysis, $\bar{L}_1$ and $\bar{L}_2$ denote the six-dataset macro-average output word counts for the two prompt conditions. We define Δ_{len} as the macro-average of the per-dataset symmetric word-count difference, $2|L_1 - L_2|/(L_1 + L_2)$. Because this quantity is computed separately within each dataset before macro averaging, it is not generally equal to the symmetric difference calculated directly from the two displayed values $\bar{L}_1$ and $\bar{L}_2$. The length-matched regime selects, for each model, the highest-flip pair among the same 28 short-answer prompt pairs subject to $\Delta_{\text{len}} \leq 10\%$.

Table 7 reveals two distinct patterns. First, the maximum pairwise flips often coincide with substantial output-length changes.

Table 6: Full-benchmark item-level official-outcome instability (flip rate %).

Model	Selection	Pair	ST-VQA	TextVQA	DocVQA	InfoVQA	ChartQA-H	ChartQA-M	Macro Avg.
InternVL3	Base vs. A*	Base-A1	8.89	8.84	2.56	14.78	14.80	8.00	9.65
	MaxFlipPair	A1-B1	10.09	9.26	3.29	16.32	15.68	8.56	10.53
Qwen2.5	Base vs. A*	Base-A1	39.61	62.44	49.67	27.99	30.96	26.96	39.61
	MaxFlipPair	A1-B2	41.04	64.84	50.01	32.13	33.92	27.44	41.56
Gemma3	Base vs. A*	Base-A2	11.44	10.92	11.09	12.39	12.16	4.64	10.44
	MaxFlipPair	A2-B1	27.26	22.30	36.60	21.89	14.64	6.16	21.48
Llama-3.2	Base vs. A*	Base-A1	67.21	75.66	83.60	66.26	31.36	40.32	60.74
	MaxFlipPair	Base-A1	67.21	75.66	83.60	66.26	31.36	40.32	60.74

Table 7: Answer-realization shifts versus output-length effects.

Model	Diagnostic regime	Pair	$\bar{L}_1$	$\bar{L}_2$	Δ_{len}	Flip
InternVL3	Max-flip pair	A1-B1	2.91	1.49	56.58%	10.53%
	Length-matched pair	A2-B1	1.61	1.49	8.60%	6.18%
	No-contract control	Base-B3	1.50	24.74	152.80%	53.45%
Qwen2.5	Max-flip pair	A1-B2	7.72	1.50	130.26%	41.56%
	Length-matched pair	Base-B1	1.42	1.30	8.58%	8.79%
	No-contract control	Base-B3	1.42	26.79	176.81%	66.37%
Gemma3	Max-flip pair	A2-B1	1.86	1.09	50.76%	21.48%
	Length-matched pair	A3-B2	1.72	1.60	7.66%	8.34%
	No-contract control	Base-B3	1.42	29.36	179.99%	61.16%
Llama-3.2	Max-flip pair	Base-A1	2.97	11.56	120.00%	60.74%
	Length-matched pair	A3-A5	8.32	8.20	9.79%	12.89%
	No-contract control	Base-B3	2.97	29.61	149.32%	65.68%

The selected max-flip pairs have Δ_{len} values ranging from 50.76% to 130.26%, and the no-contract B3 control produces still larger differences of 149.32–179.99%. The large output-length shifts indicate that prompt wording can move a model between different answer-realization modes. The resulting official flips may therefore combine changes in task behavior with changes in benchmark-contract compatibility. Second, large-scale verbosity change is not necessary for instability. After restricting the candidate set to pairs with at most 10% macro symmetric word difference, the strongest length-matched comparisons still flip 6.18% of outcomes for InternVL3, 8.79% for Qwen2.5, 8.34% for Gemma3, and 12.89% for Llama-3.2. The length-matched results provide stronger evidence that coarse output verbosity alone cannot explain the observed benchmark-facing instability.

5.4 Contract Repair (RQ4)

We design a fixed, deterministic set of CPU-only contract-repair rules. These rules include conservative wrapper removal and answer extraction, whitespace and restricted typographic-separator normalization, value-preserving whole-string numeric normalization, limited normalization of date expressions and weekday abbreviations, and the addition of predefined reference variants derived independently of model outputs and judge labels. The rules preserve numeric signs, decimal values, percentage markers, identifier-leading zeros, and the order and cardinality of multi-span answers.

We consider three variants. *Output-only* repair applies a conservative finalizer to the baseline model output. *Reference-only* repair

retains the original references and augments them with output-independent equivalent surface forms. *Bidirectional* repair applies both transformations within the same benchmark-facing comparison. We define a *break* as an originally accepted row that becomes rejected after repair, and report repair gains together with any such regressions.

Table 8 and Table 9 show that the undercount is partially recoverable through these deterministic transformations. Output-only repair yields a modest overall gain of +0.26 pp, whereas reference-only repair improves Official accuracy by +1.06 pp and recovers 1,767 Official errors without introducing any breaks. Combining both directions produces the largest improvement, +1.34 pp overall. Bidirectional repair yields 2,241 gross recoveries and introduces four breaks, corresponding to a net recovery of 2,237 rows. Of the 2,241 gross recoveries, 2,000 (89.25%) are also judged semantically acceptable by the DeepSeek semantic judge. The difference between the two one-sided variants further indicates that reference-side normalization recovers substantially more errors than output-side normalization alone. Bidirectional repair improves all six benchmarks, with the largest gains on ChartQA-Human (+2.14 pp) and ST-VQA (+1.71 pp), and improves all five contract buckets, with multi-span answers benefiting most (+3.61 pp).

Figure 3 illustrates representative output-side repairs used within the bidirectional pipeline. Although the original responses are rejected by the benchmark comparator, both the semantic judge and human annotators regard them as acceptable. The output-side rules conservatively canonicalize the model response without changing its underlying answer content, removing surface-form material that

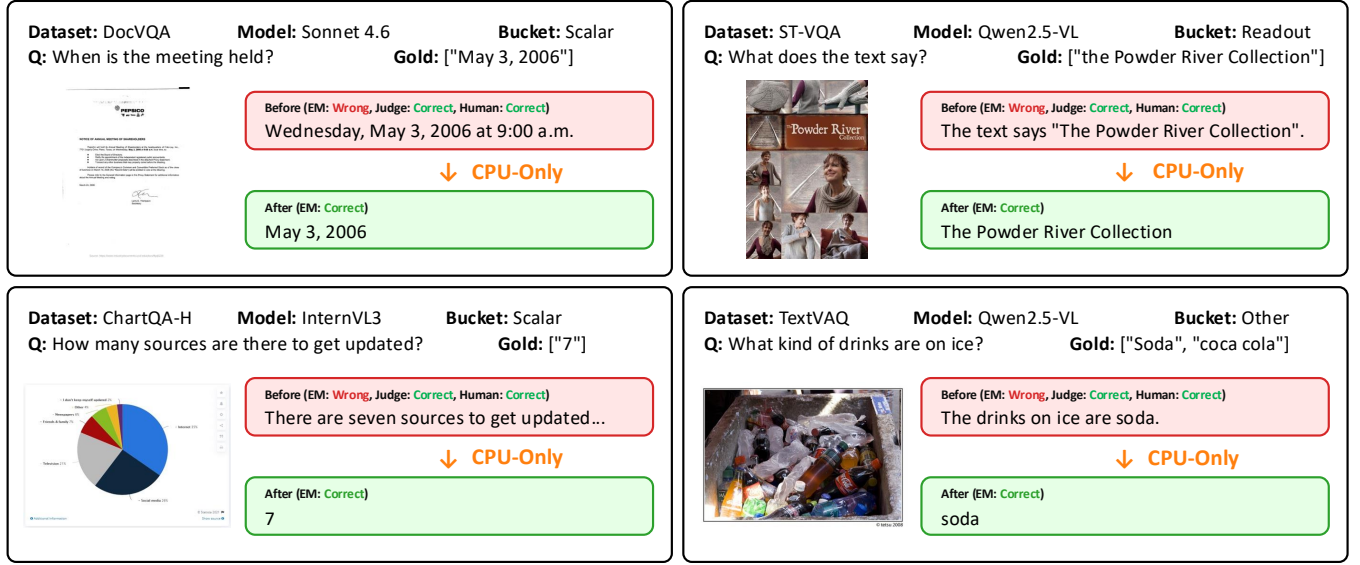


Figure 3: Representative output-side repairs inside the broader bidirectional pipeline.

Table 8: Contract repair (Δ Official) by dataset.

Dataset	Output-only	Reference-only	Bidirectional
ST-VQA	+0.35	+1.35	+1.71
TextVQA	+0.23	+0.47	+0.75
DocVQA	+0.05	+0.47	+0.52
InfoVQA	+0.07	+0.71	+0.75
ChartQA-H	+0.14	+1.58	+2.14
ChartQA-M	+0.00	+0.10	+0.10
Overall	+0.26	+1.06	+1.34

Table 9: Contract repair (Δ Official) by bucket.

Bucket	Output-only	Reference-only	Bidirectional
Scalar	+0.07	+1.31	+1.37
Readout	+0.14	+0.79	+0.94
Identifier	+0.10	+1.26	+1.36
Multi-span	-0.01	+3.24	+3.61
Other	+0.49	+0.80	+1.30
Overall	+0.26	+1.06	+1.34

interferes with benchmark matching. For example, the DocVQA response contains the correct date together with additional weekday and time information, while the repaired output retains only the reference-compatible date form.

The intervention is computationally lightweight. On a server with two Intel Xeon Gold 6148 CPUs (40 physical cores and 80 hardware threads), the full bidirectional pass over all open-weight rows completes in about 45 seconds with 32 workers, and none of the three variants requires rerunning a VLM.

6 Conclusion

Short-answer VQA scores reflect not only semantic competence but also compatibility with benchmark-specific answer contracts. Across six models and six benchmarks, semantic auditing reveals substantial undercount on text-rich datasets and systematic variation across answer forms; across four open-weight models, minor changes to short-answer instructions further induce substantial shifts in benchmark-facing outcomes. Scalar-dominated settings are markedly less affected, indicating that evaluation sensitivity is closely tied to the expected answer form. The results show that benchmark scores are shaped by the interaction among model outputs, reference representations, and evaluation rules.

Implications. Official scores remain useful, but their interpretation should account for the contracts under which answers are accepted. Differences in answer-form composition can make identical scores or score gains represent different levels of semantic success across benchmarks, while official error pools may mix genuine model failures with semantically acceptable responses rejected for surface-form reasons. Contract-aware breakdowns and targeted semantic audits can therefore complement benchmark-native metrics and make short-answer VQA evaluation more interpretable.

Limitations. Our semantic audit closely agrees with human annotations but remains imperfect, and the proposed answer-contract taxonomy is operational rather than universal. Closed-weight models are evaluated on stratified subsets, limiting fine-grained comparisons with the full open-weight evaluations. Prompt perturbations may change both answer realization and task behavior, so our conclusions concern benchmark-facing instability rather than invariance of latent semantic understanding. Finally, we focus on text-side evaluation interfaces and leave image-side and broader cross-modal perturbations to future work.

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